3.3 The Power of Convexity

WORKSHEET

The question

STEP II 2007, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Draw a curve that bends downwards, such as $y = \sin x$ on $(0, \pi)$. Mark two points on it and join them with a straight line. Is the chord above or below the curve?

STAGE 1 Checking concavity

Covers: the first part

Ask yourself Find the second derivatives of $\sin x$ and $\ln x$. What are their signs on the given intervals?

STAGE 2 Triangles

Covers: Part (i)

Pause and try Use Jensen with $f(x) = \sin x$ and the three angles A, B, C. Do they lie in an interval where sin is concave? What is their average?

STAGE 3 AM–GM in one line

Covers: Part (ii)

Pause and try AM–GM compares a product with a sum. Which concave function turns products into sums? Apply Jensen to it.

STAGE 4 Using AM–GM

Covers: Part (ii)

Ask yourself For the first inequality: the left-hand side has four terms and the right-hand side has a product of three variables. Apply AM–GM to four numbers. Why does the 16 help?

Pause and try For the minimum: you want a fifth root of $x^5y^5z^5$, so use five numbers. What should the other two be? When does equality hold?

What's really going on

Convex functions

At university the standard definition of a *convex* function (the upside-down version of concave) does not mention derivatives at all: f is convex if

$$\mathrm{f}(tx_1 + (1-t)x_2) \leq t\,\mathrm{f}(x_1) + (1-t)\,\mathrm{f}(x_2) \qquad \text{for all } x_1, x_2 \text{ and } 0 \leq t \leq 1,$$

which says exactly that chords lie above the curve. This definition makes sense for functions that are not differentiable, such as $|x|$, and for functions of many variables. The Danish engineer and mathematician Johan Jensen made convex functions a subject in their own right in 1906, with the inequality that bears his name.

Convexity everywhere

Convexity appears throughout mathematics and its applications.

- **Probability.** Jensen's inequality says that for a concave f and any random variable X, $\mathrm{E}[\mathrm{f}(X)] \leq \mathrm{f}(\mathrm{E}[X])$. In economics, with f a concave "utility" function, this explains why people prefer a certain £100 to a fifty-fifty gamble on £0 or £200, and why insurance exists.
- **Information theory.** The proof that entropy is maximised by the uniform distribution, which underlies data compression, is Jensen's inequality for ln.
- **Optimisation.** A convex function has no local minima except the global one, so "walk downhill" always finds the best answer. This is why convex optimisation problems can be solved efficiently, and why so much effort goes into recasting problems in convex form.

In Section 3.4 we will see a different side of convexity: a geometric duality between a function and its inverse.

Try this next

1. **The definition.** Parts (i) and (ii) of STEP II 2018, Question 2 (step.maths.org/questions/18-s2-q2) start from the chord definition of concavity in the box above and build up to Jensen's inequality for three points.
2. **A product of sines.** Try part (iii) of STEP II 2018, Question 2 (step.maths.org/questions/18-s2-q2).
3. **Acute triangles.** Show that $\tan x$ is convex on $(0, \frac{\pi}{2})$, and deduce that in an acute-angled triangle, $\tan A + \tan B + \tan C \geq 3\sqrt{3}$.
4. **Power means.** Show that x^2 is convex, and deduce that $\left(\frac{a_1+\cdots+a_n}{n}\right)^2 \leq \frac{a_1^2+\cdots+a_n^2}{n}$. Interpret this as a statement about the variance of a set of numbers.

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). Chapter 6 is about convexity and Jensen's inequality.
- **Claudi Alsina & Roger Nelsen, *When Less Is More: Visualizing Basic Inequalities*** (Mathematical Association of America). Inequalities proved by pictures, many of them using convexity.
- **Stephen Boyd & Lieven Vandenberghe, *Convex Optimization*** (Cambridge University Press). A modern classic, free online; the first chapters show why convexity matters in practice.

The full guided solution is in Section 3.3:
https://beyondstep.keithmcnulty.org/ch03/convexity.html